



1.1 Arithmetic and geometric sequences and series

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

An arithmetic sequence has constant common difference d , with $u_n = u_1 + (n-1)d$ and sum $S_n = \frac{n}{2}(2u_1 + (n-1)d)$. A geometric sequence has constant common ratio r , with $u_n = u_1 \cdot r^{(n-1)}$ and sum $S_n = \frac{u_1(r^n - 1)}{r - 1}$ for $r \neq 1$. If $|r| < 1$, the infinite geometric series converges to $S_\infty = \frac{u_1}{1-r}$.

Worked example: follow the reasoning

An AP has $u_1=5$, $d=3$. Find S_{20} .
 $S_{20} = \frac{20}{2}(2(5) + 19(3)) = 10(10 + 57) = 10(67) = 670$.

Quick check · multiple choice

A geometric sequence has $u_1=4$ and $r=1/2$. What is S_∞ ?

- A 4
- B 6
- C 8
- D Infinite (does not converge)

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 An arithmetic sequence has $u_3=11$ and $u_7=27$. [7]
- (a) Find d and u_1 .
 - (b) Find S_{15} .
- A geometric sequence has the same u_1 and $r=1.5$.
- (c) Find u_5 of the geometric sequence.

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) $u_7 - u_3 = 4d = 16$ so $d = 4$; $u_1 = 11 - 2d = 3$.
(b) $S_{15} = 15/2(2(3) + 14(4)) = 15/2(6 + 56) = 15/2(62) = 465$.
(c) $u_5 = u_1 \cdot r^4 = 3(1.5)^4 = 3(5.0625) = 15.1875$.

Quick check answer

C. $S_{\infty} = u_1/(1-r) = 4/(1-0.5) = 8$, since $|r| = 0.5 < 1$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.